ALEXIS ROBIN



AI · DATA SCIENCE · SOFTWARE ENGINEERING

Master's student · University of Lille × Centrale Lille Institut

Agentic systems · Scientific data · Open source and software security

Internship: 22 March–21 August 2027 · International mobility

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Master's student in AI for healthcare. Built an agentic genomics backend at CIRAD and a DOI-archived ML benchmark; independently reported a memory-safety vulnerability in Tokio.

PROFESSIONAL EXPERIENCE

CIRAD · Agentic AI and Data Engineering Intern | Apr–Aug 2026

- Built the Python LLM agent and FastAPI backend of GenomInt, enabling natural-language exploration of a Neo4j pangenomic knowledge graph using tools and hybrid retrieval.
- Implemented server-side graph filters, paginated APIs and batch FASTA exports; enforced read-only Cypher validation, authorization, timeouts and rate limiting.
- Containerized and deployed services with Podman, Helm and Kubernetes (K3s); coordinated backend/frontend integration and API contracts.

SMAG · Full-Stack Development Intern | Feb–May 2025

- Developed a FastAPI/Vue.js application automating agricultural CSV imports, joins and exports for digital-agriculture teams.

OPEN SOURCE AND SECURITY RESEARCH

Tokio · Independent vulnerability discovery | Oct 2026

Independently reported a use-after-free race in Tokio 1.54.0's `Interval` on the multithreaded runtime (CVE-2026-115567). Tokio powers Rust async systems and is used in Google, AWS and Microsoft projects. Credited reporter; after production crashes, v1.54.0 was yanked and fixed in 1.54.1.

SELECTED PROJECTS

HealthGraphBench Independent ML research · DOI-archived v0.1

Built a temporal benchmark comparing tabular, heuristic and graph ML on FDA MAUDE/CMS data. On exploratory MAUDE results, neighbor frequency beat GraphSAGE (Recall@10: 0.192 vs 0.173); v0.2 extends to Medicare Part D.

Sillage Rust · Document search and Linux systems

Contributing to a Rust/Linux launcher and source-attributed document search (Tantivy, SQLite; lexical and optional semantic retrieval).

Nacre Rust · Compilers

Developing a Rust compiler for a Java-like language, emitting C11 or optionally LLVM IR.

Rature Project initiator · Early development

Initiated a pre-release open-source knowledge project for people and AI agents; designing source-anchored research, a draft protocol and Rust/PostgreSQL architecture.

EDUCATION

University of Lille · Centrale Lille Institut | 2026–2028 (expected)

Master's in Health Engineering — AI Management in Healthcare (MIAS).

University of Montpellier · IUT Montpellier-Sète | 2023–2026

Bachelor of Technology (BUT) in Computer Science · Secure Networked Application Deployment track.

TECHNICAL SKILLS

Programming: Python, Rust, SQL, TypeScript, Java, C
AI and data: PyTorch, Transformers, scikit-learn, pandas; LLM agents, RAG, benchmarking
Backend and systems: FastAPI, Neo4j/Cypher, PostgreSQL, SQLite, Linux, Docker, K3s
Security: Concurrency analysis, authorization, responsible vulnerability disclosure
Languages: French (native); English